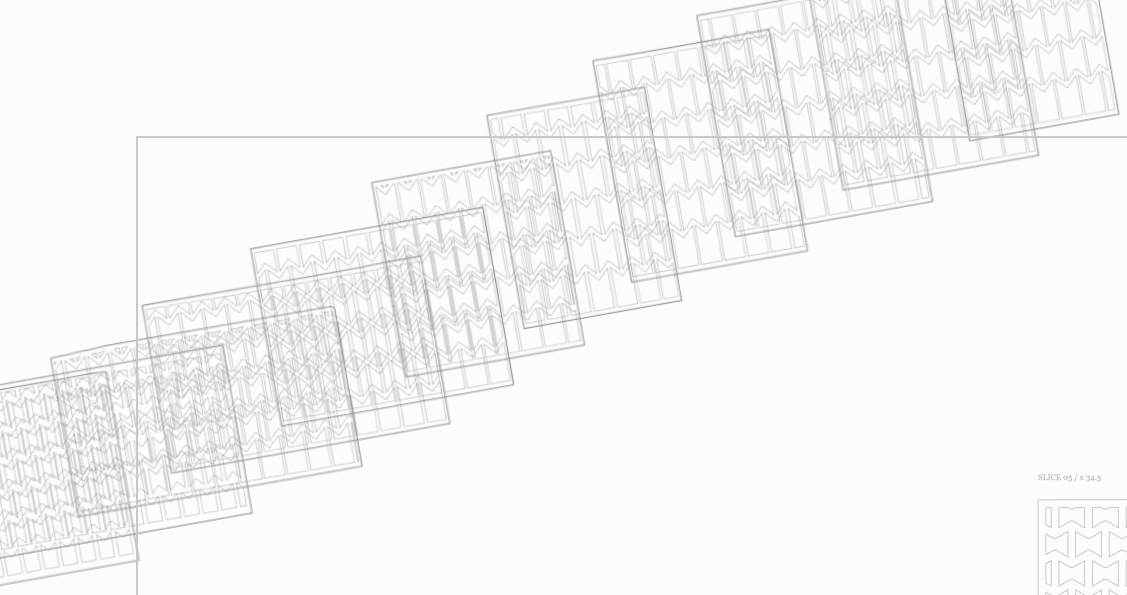
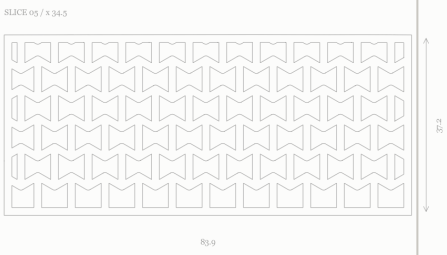
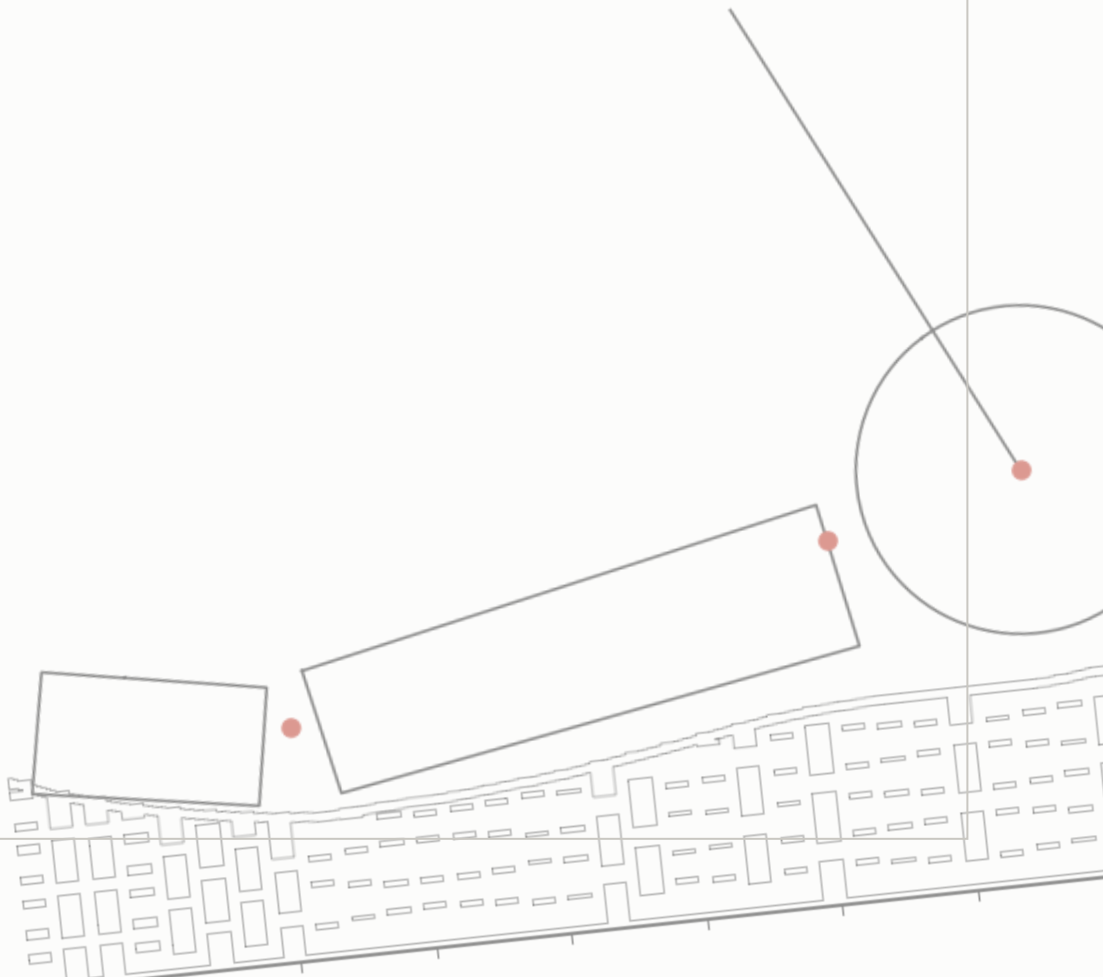




RE-ENTRANT AUXETICS

FROM CELL MODELS TO A RUNNING MIDSOLE

MOS PROJECT 2026–27 · RE-ENTRANT HONEYCOMB STRAND



ABSTRACT

Re-entrant honeycombs are the archetypal auxetic lattice: under compression their bow-tie cells fold inward, giving a negative Poisson's ratio. This report compares four analytical descriptions of the cell — Gibson & Ashby, Masters & Evans, Malek & Gibson and Ohtaki — against plane-stress finite-element simulations in ANSYS, and then applies the cell to a running midsole. Measured running data (Fukuchi et al.) drive a three-segment pseudo-foot that loads 34 lattice slices; each slice is solved as a beam on a lattice foundation, and the lattice is chosen per zone of the foot.

At a wall thickness ratio of t/l = 0.4 the finite-element cell is 1.2–2.9 times stiffer and markedly less auxetic (ν_{12} = −0.14, ν_{21} = −0.33) than the models predict, and it buckles at 0.36 MPa. Ohtaki's large-deflection model follows the simulation closely up to that point but cannot predict the collapse. In the midsole, forefoot strains reach about 20 % while midfoot and heel strains are governed by the foot's shape rather than its load.

CONTENTS

I Re-entrant auxetics 3
II Four analytical models 5
III Finite-element comparison 7
IV Application: running footwear 10
V Midsole optimisation 11
VI Discussion and outlook 17
— References 18

Dashed boxes marked PLACEHOLDER are reserved for ANSYS screenshots, photographs and drawings still to be added.

I

RE-ENTRANT AUXETICS

Materials that get thinner when compressed — and fatter when stretched

1.1 Negative Poisson's ratio

Poisson's ratio **ν** measures how a material changes width when loaded along its length. Almost every conventional material has **ν** between 0 and 0.5: a rubber band narrows as it is stretched, and a foam block bulges sideways when squeezed. *Auxetic* materials have **ν** < 0; they widen when stretched and draw inward when compressed. The behaviour was demonstrated in re-entrant polymer foams by Lakes [1] and named by Evans et al. [2].

Auxetic behaviour is a property of geometry rather than chemistry. It appears in structures whose internal members rotate or fold as the structure deforms: re-entrant honeycombs, chiral lattices, rotating squares and related mechanisms [3]. The consequences are practical — higher indentation resistance, higher shear stiffness relative to bulk stiffness, energy absorption and the ability to curve into a dome (synclastic curvature) rather than a saddle.

1.2 The re-entrant honeycomb

A conventional hexagonal honeycomb has inclined walls that point outward. Turning the wall angle θ negative makes the inclined walls point back into the cell, producing the bow-tie shape of the re-entrant honeycomb. Under vertical compression these walls rotate further inward and pull the vertical walls together, so the whole lattice contracts sideways.

The cell is described by four lengths and an angle: the vertical wall h, the inclined wall l, the wall thickness t, the depth b and the angle θ (negative for re-entrant). The two ratios h/l and t/l, together with θ, set the in-plane properties; the solid's modulus E_s scales them.

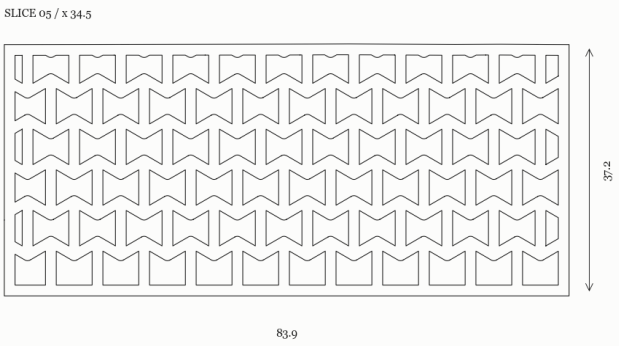


Fig. 1 A lattice slice from the midsole of Chapter V, showing the re-entrant cell clipped to the foot section.

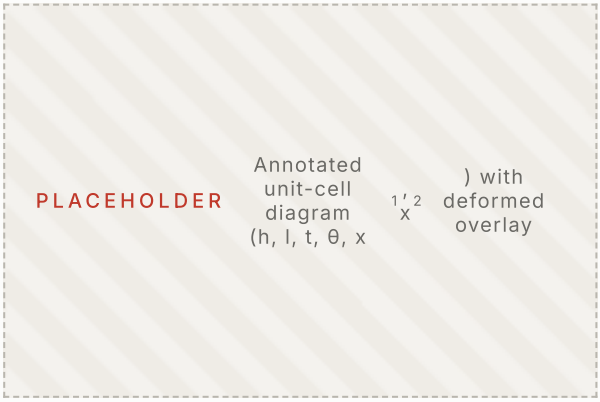


Fig. 2 Unit cell and notation.

1.3 Scope of this strand

This report is one of four strands of a team project comparing auxetic and tensegrity concepts for footwear: re-entrant honeycombs (this report), chiral lattices, rotating squares and tensegrity structures. It asks two questions.

- How well do the standard analytical models describe a re-entrant cell of printable proportions, judged against finite-element simulation?
- When the cell is placed in a running-shoe midsole and loaded by a realistic stance, how does it deform, and which cell belongs where?

Throughout, the solid is taken as a thermoplastic polyurethane with E_s = 19.3 MPa and ν_s = 0.45. The printed samples (TPU 90A or 95A) will fix the true value; all stiffnesses in this report scale linearly with it.

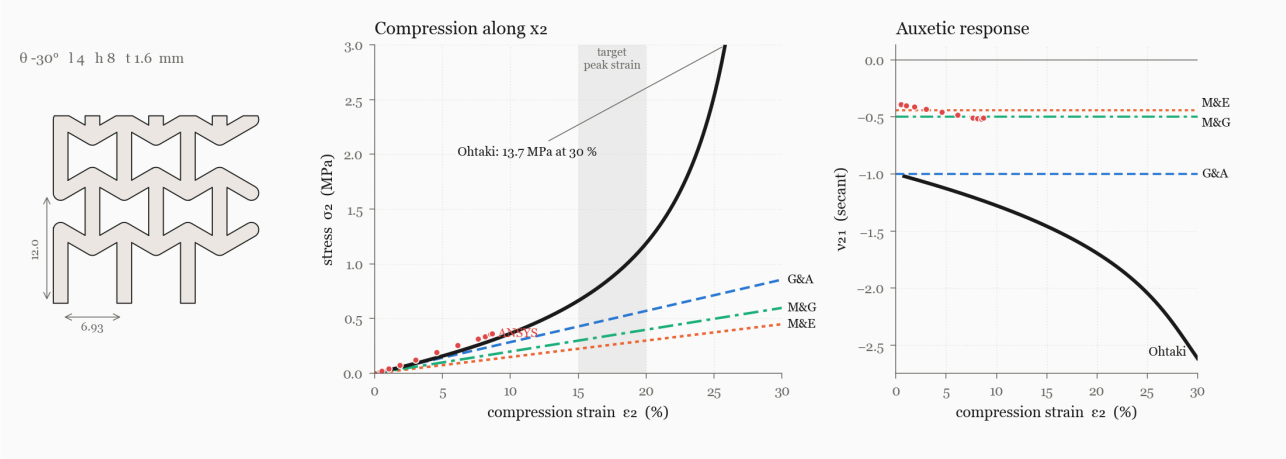


Fig. 3 Preview of Chapter III: the forefoot cell (left), its compression response (centre) and Poisson's ratio (right) from four models and ANSYS.

PLACEHOLDER Photograph of printed TPU re-entrant sample (front view, with scale)

Fig. 4 Printed re-entrant sample.

II

FOUR ANALYTICAL MODELS

Gibson & Ashby · Masters & Evans · Malek & Gibson · Ohtaki

All four models treat the cell walls as beams and differ in which deformation mechanisms they keep. The first three are linear (small strain) and share one compliance form; the fourth follows the walls through large rotations.

2.1 A common compliance form

Each inclined wall deforms perpendicular to its axis (bending, hinging, shear) with compliance C_p and along its axis (stretching) with compliance C_a; the vertical wall stretches under x_2 load with compliance C_v. In units of $l^3 / (E_s\, b\, t^3)$, with $\alpha = h/l$ and $\beta = t/l$:

$$D_1 = \sin^2\theta\, C_p + \cos^2\theta\, C_a, \quad D_2 = \cos^2\theta\, C_p + \sin^2\theta\, C_a + C_v \tag{1}$$

$$E_1 = E_s\, \beta^3 \cos\theta \,/\, [(\alpha + \sin\theta)\, D_1], \quad E_2 = E_s\, \beta^3 (\alpha + \sin\theta) \,/\, [\cos\theta\, D_2] \tag{2}$$

$$\nu_{12} = \sin\theta \cos^2\theta\, (C_p - C_a) \,/\, [(\alpha + \sin\theta)\, D_1], \quad \nu_{21} = \sin\theta\, (\alpha + \sin\theta)(C_p - C_a) \,/\, D_2 \tag{3}$$

Gibson & Ashby [4]

Walls bend only: $C_p = 1$, $C_a = C_v = 0$. The model is the stiffest of the linear three in Poisson's ratio and, for $\theta = -30°$ with $h/l = 2$, gives exactly $\nu = -1$ and $E_1 = E_2$.

Masters & Evans [5]

Adds axial stretching of the walls, $C_a = \beta^2$, $C_v = 2\alpha\beta^2$, and optionally a hinge compliance at the joints, $C_p = 1 + K_f/K_h$. Stretching softens the cell and weakens the auxetic effect; hinging is not used here (rigid joints).

Malek & Gibson [6]

Recognises that thick walls meet in solid joints, so only the shorter lengths $l_b = l - t/(2\cos\theta)$ and $h_b = h - t(1 - \sin\theta)/\cos\theta$ bend, and adds Timoshenko shear: $C_p = 1 + (2.4 + 1.5\nu_s)\,(t/l_b)^2$. The paper validates $\theta > 0$ only; negative θ is an extrapolation.

Ohtaki [7]

Ohtaki et al. treat each half inclined wall as an *elastica*: an inextensible beam allowed large rotations. With $\varphi(s)$ the wall angle along its length and forces non-dimensionalised by $E_s I/(l/2)^2$,

$$\varphi'' = f_x \sin\varphi - f_y \cos\varphi, \quad \varphi(0) = \theta, \quad \varphi'(1) = 0 \tag{4}$$

which is solved by shooting for each load step. The result is a nonlinear stress–strain curve and a Poisson's ratio that changes with strain. Its first load step reproduces Gibson & Ashby, which was used to verify the implementation. In tension along x_1 the walls straighten and the strain saturates at $1/\cos\theta - 1$; in x_2 compression the curve stiffens sharply past about 20 % as the walls lock, since the model ignores wall contact.

2.2 Predictions for the two midsole cells

Chapter V selects two cells: a stiffer forefoot cell (θ −30°, l 4, h 8, t 1.6 mm; t/l = 0.40) and a softer midfoot/heel cell (θ −30°, l 5, h 10, t 1.0 mm; t/l = 0.20). Figure 5 compares the four models for both.

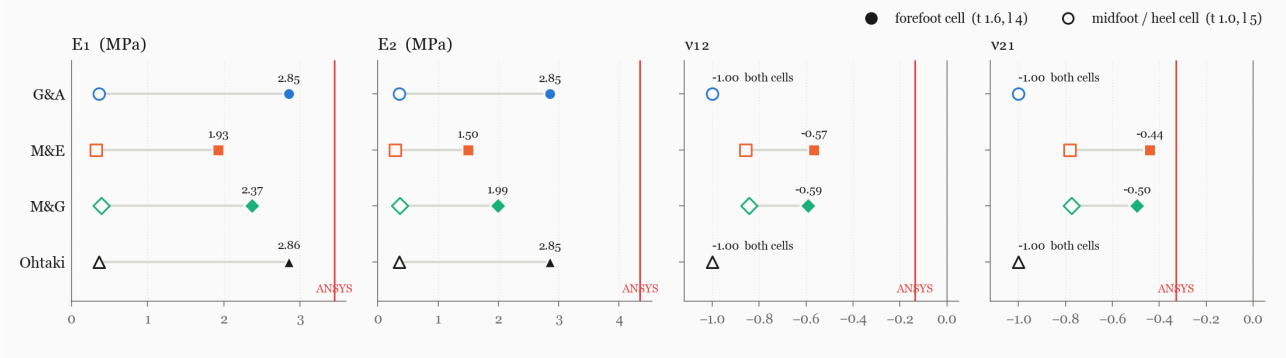


Fig. 5 Small-strain in-plane properties. Filled: forefoot cell; open: midfoot/heel cell; red line: ANSYS (forefoot cell, Chapter III).

Forefoot cell	E_1 (MPa)	E_2 (MPa)	ν_{12}	ν_{21}
Gibson & Ashby	2.85	2.85	−1.00	−1.00
Masters & Evans	1.93	1.50	−0.57	−0.44
Malek & Gibson	2.37	1.99	−0.59	−0.50
Ohtaki (first step)	2.86	2.85	−1.00	−1.00

The models span a factor of two in stiffness and a factor of two in Poisson's ratio for the same cell. Adding mechanisms (stretching, shear) softens the cell and weakens the auxetic effect; shortening the bending length (rigid joints) stiffens it again. Which of these dominates for a thick printed wall is the question for the finite-element comparison.

III

FINITE-ELEMENT COMPARISON

A 4 × 4 block of the forefoot cell in ANSYS Mechanical

3.1 Model

The forefoot cell was modelled as a 4 × 4 block, 27.7 mm wide, with the right edge closed and the top mirrored from the bottom so the block is 180° symmetric; walls span 56 mm in height. Solid strips 1.6 mm thick carry the load into the lattice (top and bottom for x_2 loading, left and right for x_1). The geometry was generated as an exact outline and analysed as a 2D plane-stress body, 5 mm thick, meshed with quadrilaterals at about 0.3 mm (four or more elements through each wall).

- **Run A** — x_2 compression to 1 %, linear elastic (E_s 19.3 MPa, ν_s 0.45).
- **Run B** — x_1 tension to 1 %, same material.
- **Run C** — x_2 compression to 30 %, Neo-Hookean (μ = 6.66 MPa, D_1 = 0.0311 MPa^{-1}), large deflection.

Strains were measured on the middle 2 × 2 cells, away from the strips, from probe displacements at four wall joints: ε_2 over a 12 mm cell pitch and ε_1 over 6.93 mm. Stress is the strip reaction divided by the loaded cross-section.

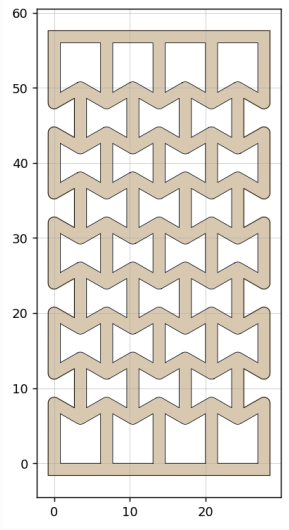


Fig. 6 Block geometry for runs A and C (strips top and bottom).

PLACEHOLDER ANSYS mesh close-up with probe locations P1–P4

Fig. 7 Mesh and probe joints.

3.2 Small-strain properties

Forefoot cell	E_1 (MPa)	E_2 (MPa)	ν_{12}	ν_{21}
ANSYS	3.45	4.35	−0.14	−0.33
Gibson & Ashby	2.85 (−17 %)	2.85 (−34 %)	−1.00	−1.00
Masters & Evans	1.93 (−44 %)	1.50 (−66 %)	−0.57	−0.44
Malek & Gibson	2.37 (−31 %)	1.99 (−54 %)	−0.59	−0.50

Every model under-predicts the stiffness, Gibson & Ashby least, and every model over-predicts the auxetic effect, most of all sideways (ν_{12}). Three effects explain this. At t/l = 0.4 the solid joints occupy a large share of each wall, so less of it bends — Malek & Gibson capture part of this through l_b, but not enough. Thick walls also shear and compress in ways the beam models only partly include. Finally, the bonded strips restrain the end rows from contracting sideways, and in an auxetic block this restraint stiffens the interior as well: the middle cells saw only half of the applied strain.

The two runs are not fully consistent with a homogeneous orthotropic material ($\nu_{12}/E_1 = -0.040$ against $\nu_{21}/E_2 = -0.076$). A 4 × 4 block is too small to be free of its boundaries; a larger block or periodic boundary conditions would tighten the comparison.

PLACEHOLDER Run A — deformed shape, vertical displacement contour

Fig. 8 Run A, x_2 compression at 1 %.

PLACEHOLDER Run B — deformed shape, horizontal displacement contour

Fig. 9 Run B, x_1 tension at 1 %.

PLACEHOLDER Equivalent stress contour, run A — showing joint stress concentration

Fig. 10 Stress concentrates at the joints, where the thick walls meet.

3.3 Large strain and collapse

In run C the stress rises almost linearly to a peak of **0.36 MPa** at 8.7 % strain in the middle cells (17 % across the whole block), then falls. Past the peak the middle cells unload slightly while other rows continue to collapse: the deformation localises and the cells buckle, the classic onset of a honeycomb's collapse plateau. The solution stopped converging at 20 % overall strain through excessive element distortion, by which point collapse was well developed.

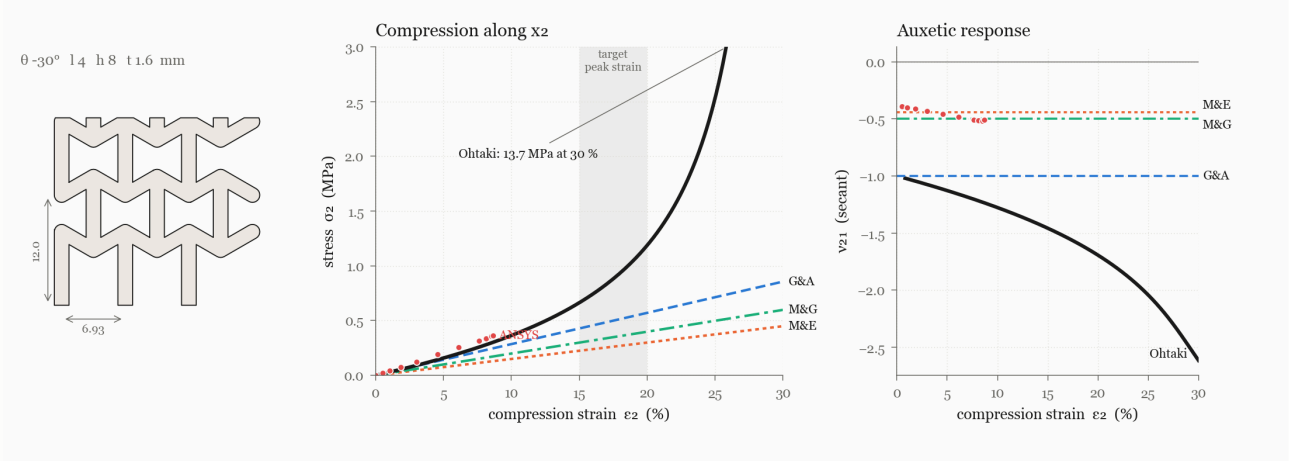


Fig. 11 Forefoot cell under x_2 compression. ANSYS points (red) are shown up to the load peak; the shaded band is the strain targeted in the midsole optimisation.

Up to the peak, ANSYS follows **Ohtaki** closely and lies above the three linear models, which fall further behind as strain grows. Ohtaki's steep stiffening past 20 % never occurs: the real cell buckles first. Poisson's ratio grows more negative with strain, from −0.40 to −0.51 — Ohtaki's trend, but much milder, and close to the Masters & Evans and Malek & Gibson values.

PLACEHOLDER Run C at 0 %

Fig. 12a

PLACEHOLDER Run C at peak load (≈17 %)

Fig. 12b

PLACEHOLDER Run C after collapse (≈20 %)

Fig. 12c

For design, the forefoot cell therefore offers about 0.36 MPa before it collapses. Peak forefoot pressures in running are of a similar order, so collapse is within reach — a feature if a controlled plateau is wanted for cushioning, a limit if it is not.

IV

APPLICATION: RUNNING FOOTWEAR

Why put an auxetic lattice under the foot?

A running midsole must absorb impact, return energy and stay stable as the load moves from heel or forefoot to the toes in about a fifth of a second. Conventional foams spread outward under load. An auxetic structure instead gathers material under the load, increasing local stiffness where the foot presses (indentation resistance), and curves synclastically, so it can cup the foot rather than flatten beneath it. These properties are repeatedly cited as motivation for auxetics in protective equipment and footwear [3, 8].

4.1 Existing work

Reviews of auxetic metamaterials list sports footwear among the most promising applications, alongside protective padding and impact absorbers [3, 8]. Additively manufactured lattice midsoles are now commercial, which makes graded, printed cells a realistic manufacturing route; most commercial lattices are, however, conventional rather than auxetic. *[Add footwear-specific auxetic studies here — e.g. printed re-entrant or chiral insoles and midsoles, plantar-pressure and comfort studies.]*

4.2 Running mechanics

Ground reaction force in running peaks at two to three times body weight. The centre of pressure travels from the heel (or midfoot, for forefoot strikers) to the metatarsal heads and toes, and the metatarsophalangeal (MTP) joint dorsiflexes by roughly 30–40° before toe-off [9, 10]. A midsole therefore sees a moving, uneven load and a bending demand concentrated behind the toes. Chapter V uses one measured stance from the public running dataset of Fukuchi et al. [11] to reproduce this load.

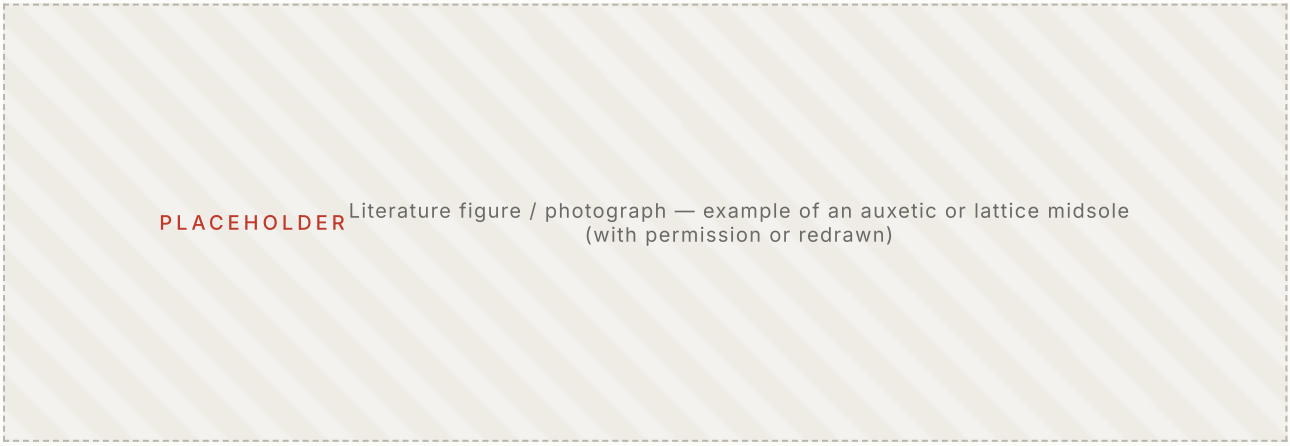


Fig. 13 Precedent: printed lattice midsoles.

V

MIDSOLE OPTIMISATION

Filling a midsole, loading it with a running stance, choosing a cell per zone

5.1 Filling the midsole

A bulk midsole volume (flat bottom, curved footbed, 270 mm long) was cut into 34 slices across the foot, 5 mm thick with 3 mm gaps. Each slice is filled with the re-entrant lattice and closed by a skin wall; a flat bottom plate and a top plate following the footbed join the slices into a single watertight body (242 cm^3). Because the slices lie across the foot, the lattice plane is the y–z plane and its x_2 axis is vertical — the direction of the foot load.

PLACEHOLDER CAD screenshot — bulk midsole volume (midsole_base.step)

Fig. 14 Bulk midsole volume.

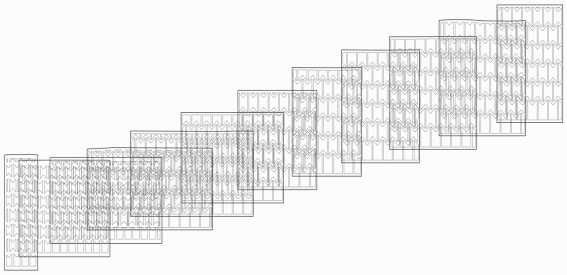


Fig. 15 Slice stack, every third slice shown.

5.2 The pseudo-foot

The foot is represented by three rigid bodies — a heel sphere, a midfoot block and a toe block — joined by hinges along y at the ankle, between heel and midfoot (three-quarters up the midfoot's rear face) and at the MTP (60 % up the gap between midfoot and toes). The heel and midfoot are kept rigid together, since the dataset has no midfoot marker.

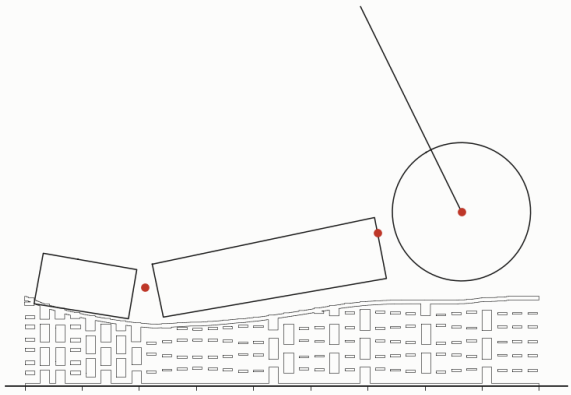


Fig. 16 Pseudo-foot at mid-stance over the midsole section (y = 60 mm). Red: hinges.

5.3 From running data to foot motion

One right-foot stance (213 ms, peak 1523 N) of subject 1 at 3.5 m/s was taken from the Fukuchi dataset. The vertical force and the centre of pressure, projected on the heel–metatarsal line, split the load into heel and forefoot shares. The heel and metatarsal markers give the foot's pitch and the shank markers the ankle angle, both relative to standing; the MTP bend follows Stefanyshyn & Nigg [9]. All signals are smoothed with a zero-lag exponential moving average. This runner lands on the forefoot, so the heel carries little load.

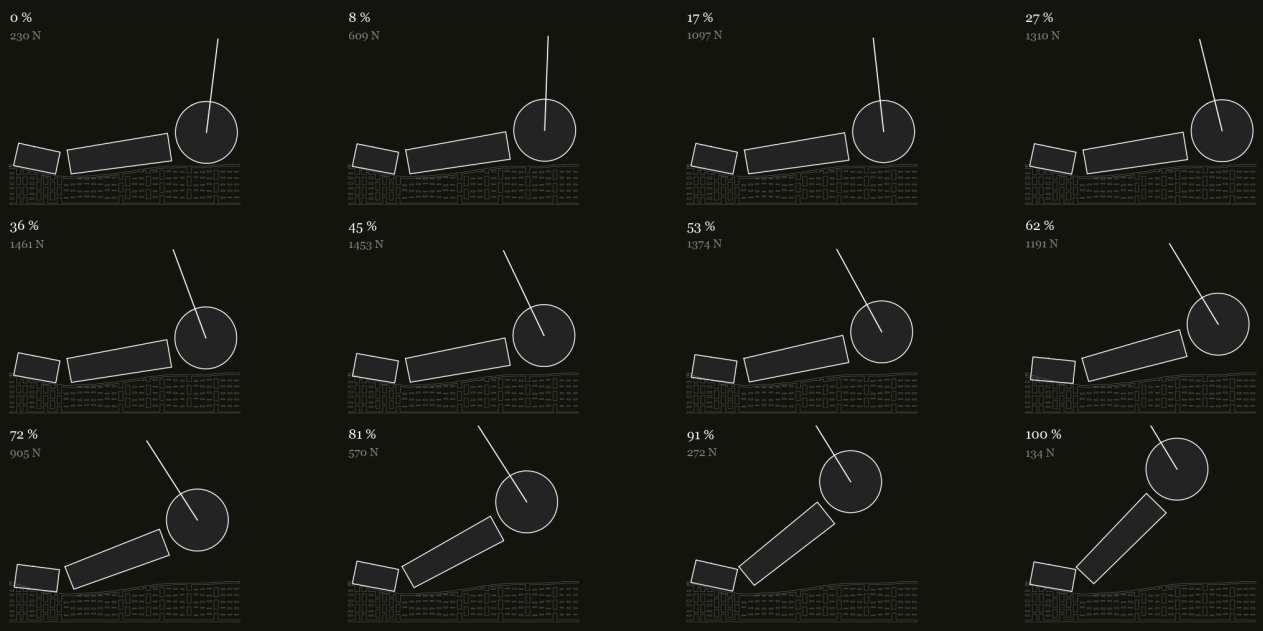


Fig. 17 Twelve frames of the stance: pseudo-foot posed from the measured angles.

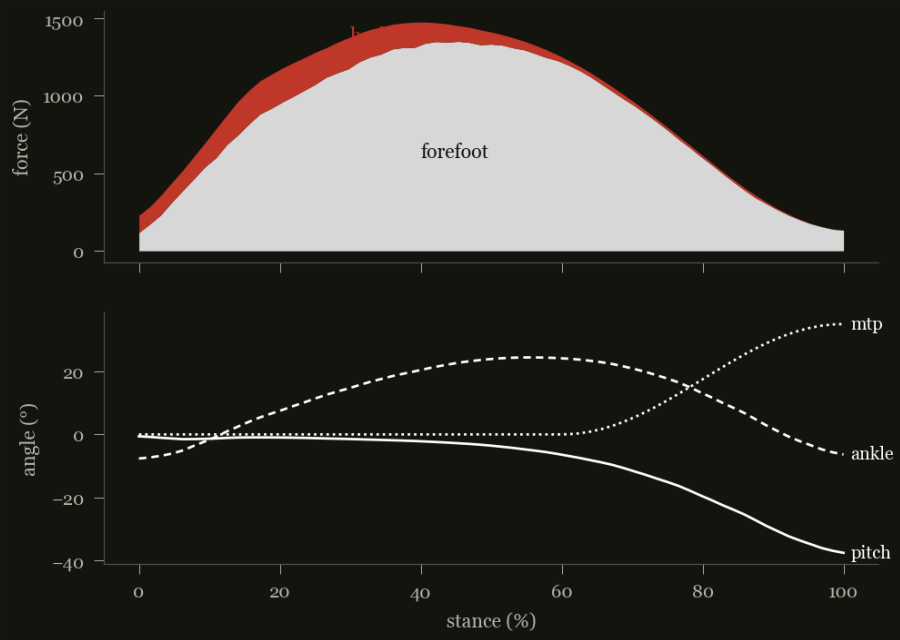


Fig. 18 Force split (white forefoot, red heel) and joint angles.

5.4 Loading the slices: a beam on a lattice foundation

The posed foot is lowered into the midsole by a *sink*. For each slice, the top plate and skin are treated as a beam across the foot resting on a Winkler foundation of lattice columns. Each column's stress follows Ohtaki's compression curve at its strain, and the rigid foot acts as an indenter wherever it would cut into the top surface. The sink is then solved for every frame so that the summed slice forces equal the measured ground reaction.

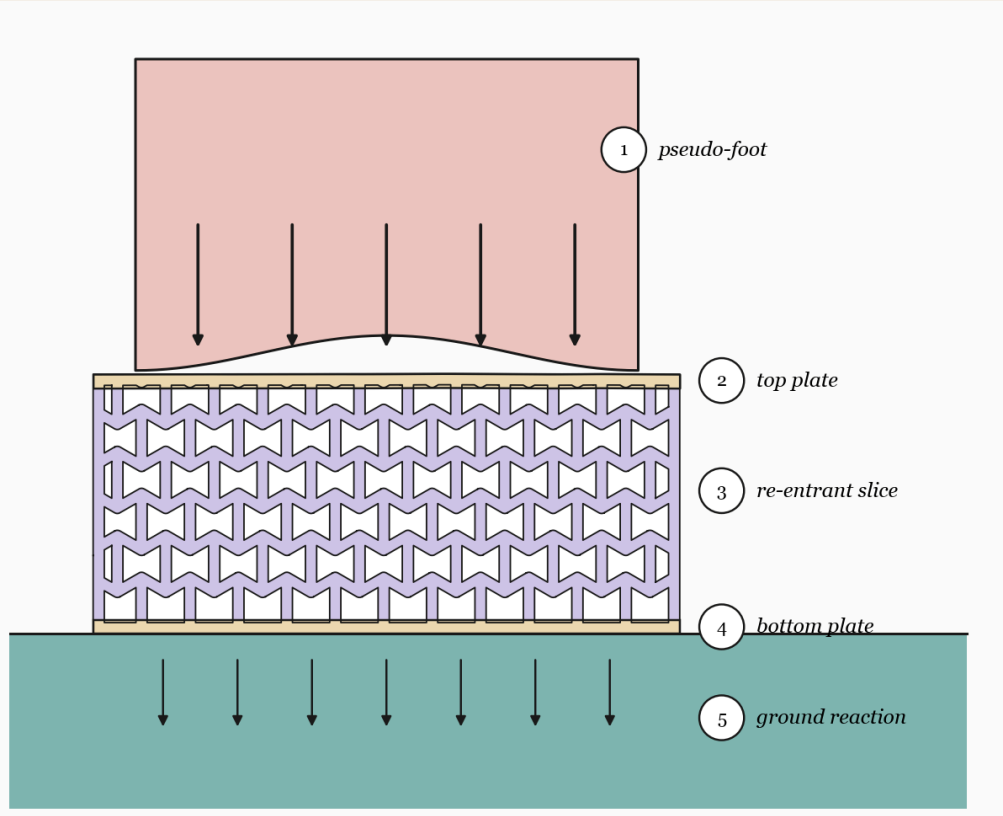


Fig. 19 Load path across the foot at a forefoot slice: (1) pseudo-foot, (2) top plate, (3) re-entrant slice, (4) bottom plate, (5) ground reaction.

The foundation idea assumes the slices interact only through the plates and that neighbouring columns do not shear against each other. Chapter III suggests the Ohtaki law is reasonable up to buckling, so strains beyond about 17 % (overall) should be read with caution.

5.5 Zone search

The midsole was divided at the hinge lines into forefoot, midfoot and heel. For each zone, 200 candidate cells (θ −10° to −30°, l 3.5–7 mm, h/l 1–2, t 1.0–2.6 mm, with t/l ≤ 0.4, a clear bow-tie gap ≥ 2 mm and a cell no taller than 16 mm) were ranked by Masters & Evans E_2. The full stance was repeated, moving each zone toward a peak column strain of 17.5 % and preferring the most auxetic cell among near ties.

Zone	θ	l	h	t (mm)	E_2 (MPa)	ν_{21}	Peak strain
Forefoot	−30°	4	8	1.6	1.50	−0.44	19.8 %
Midfoot	−30°	5	10	1.0	0.29	−0.78	10.5 %
Heel	−30°	5	10	1.0	0.29	−0.78	8.4 %

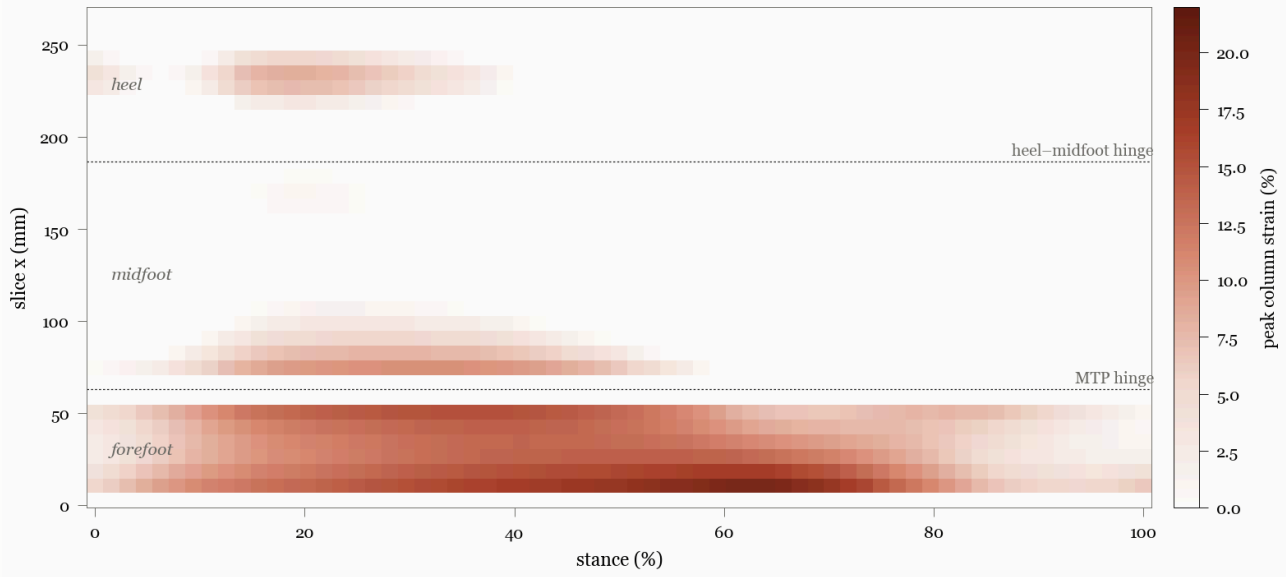


Fig. 20 Peak column strain per slice through the stance (zoned lattice). Dashed: hinge lines.

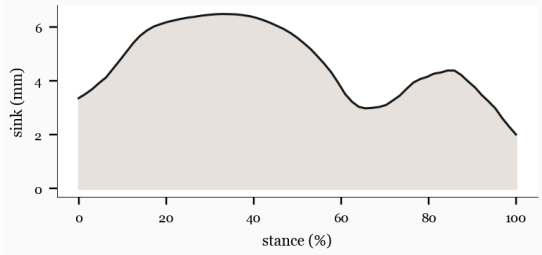


Fig. 21 Sink of the foot into the sole, solved per frame.

Strain concentrates under the metatarsal heads. In the midfoot and heel, strain is set by the foot's shape sinking into the sole rather than by the force there, so softer cells barely change it; the search therefore settles on the softest, most auxetic cell allowed. This holds for a forefoot striker only: a heel striker would need a stiffer heel.

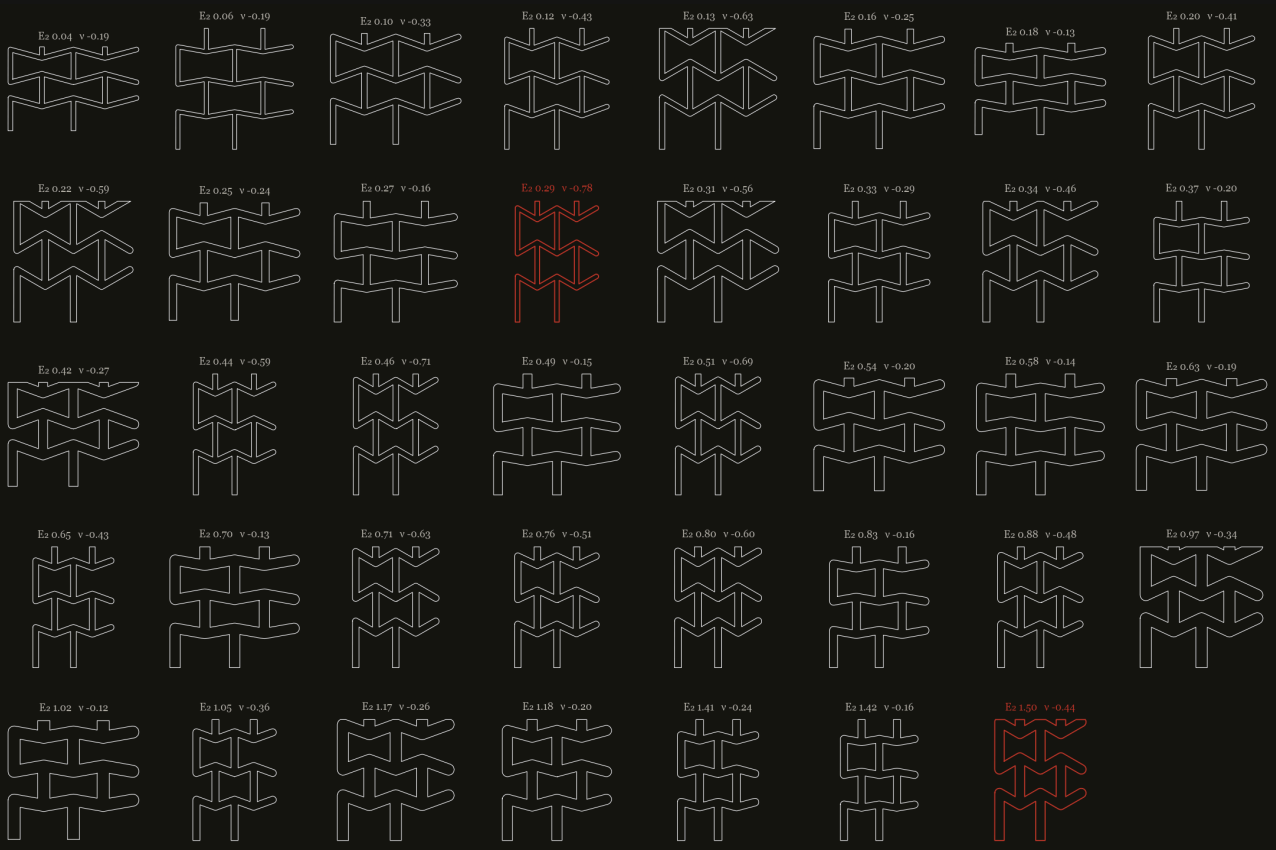


Fig. 22 Forty of the candidate cells, softest to stiffest by Masters & Evans E_2. Red: cells chosen for the forefoot and the midfoot/heel.

The chapter III results revise this picture. At t/l = 0.4 the chosen forefoot cell is about 2.9 times stiffer than the Masters & Evans value used in the search, so the real forefoot strains would sit below the predicted 19.8 % — and below the buckling strain. A second search using an ANSYS-calibrated stiffness, or a thinner wall (t/l ≈ 0.25–0.3) that stays closer to beam theory and keeps more of the auxetic effect, are the natural next steps.

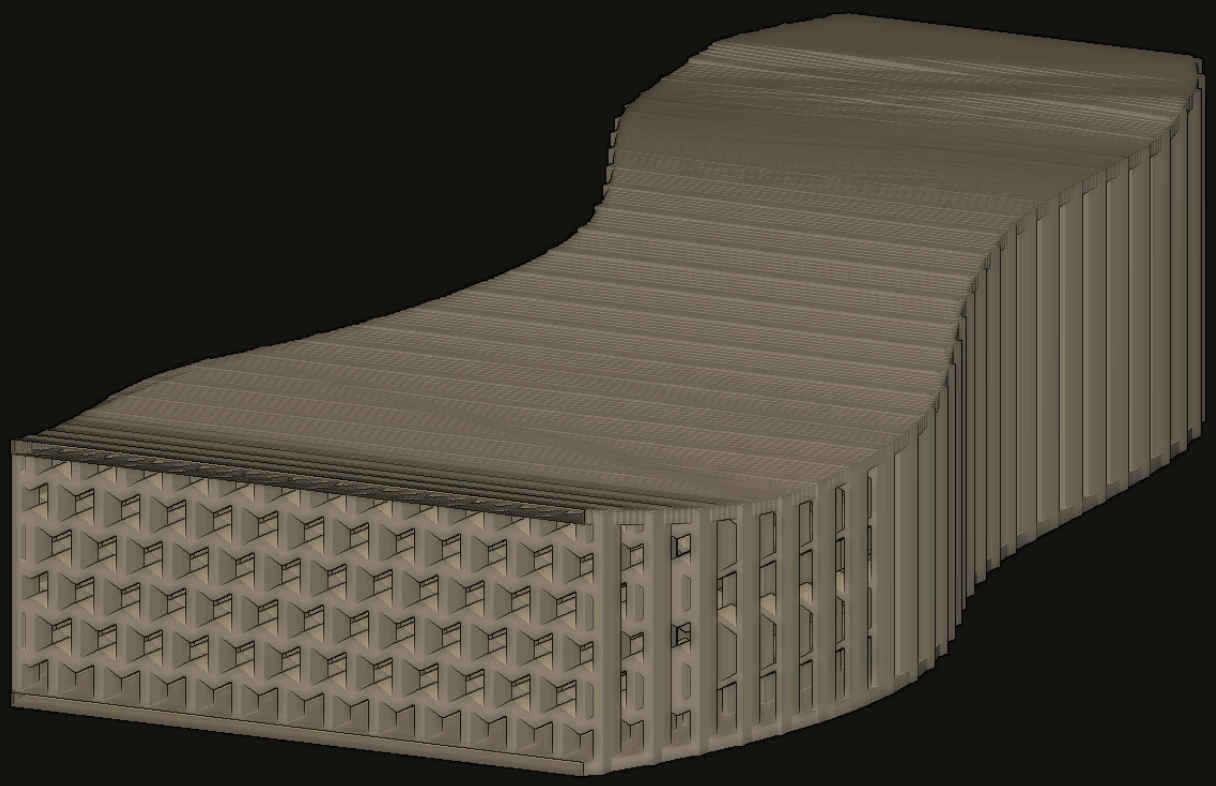


Fig. 23 Zoned midsole cut across the foot at x = 38 mm, through the forefoot lattice.

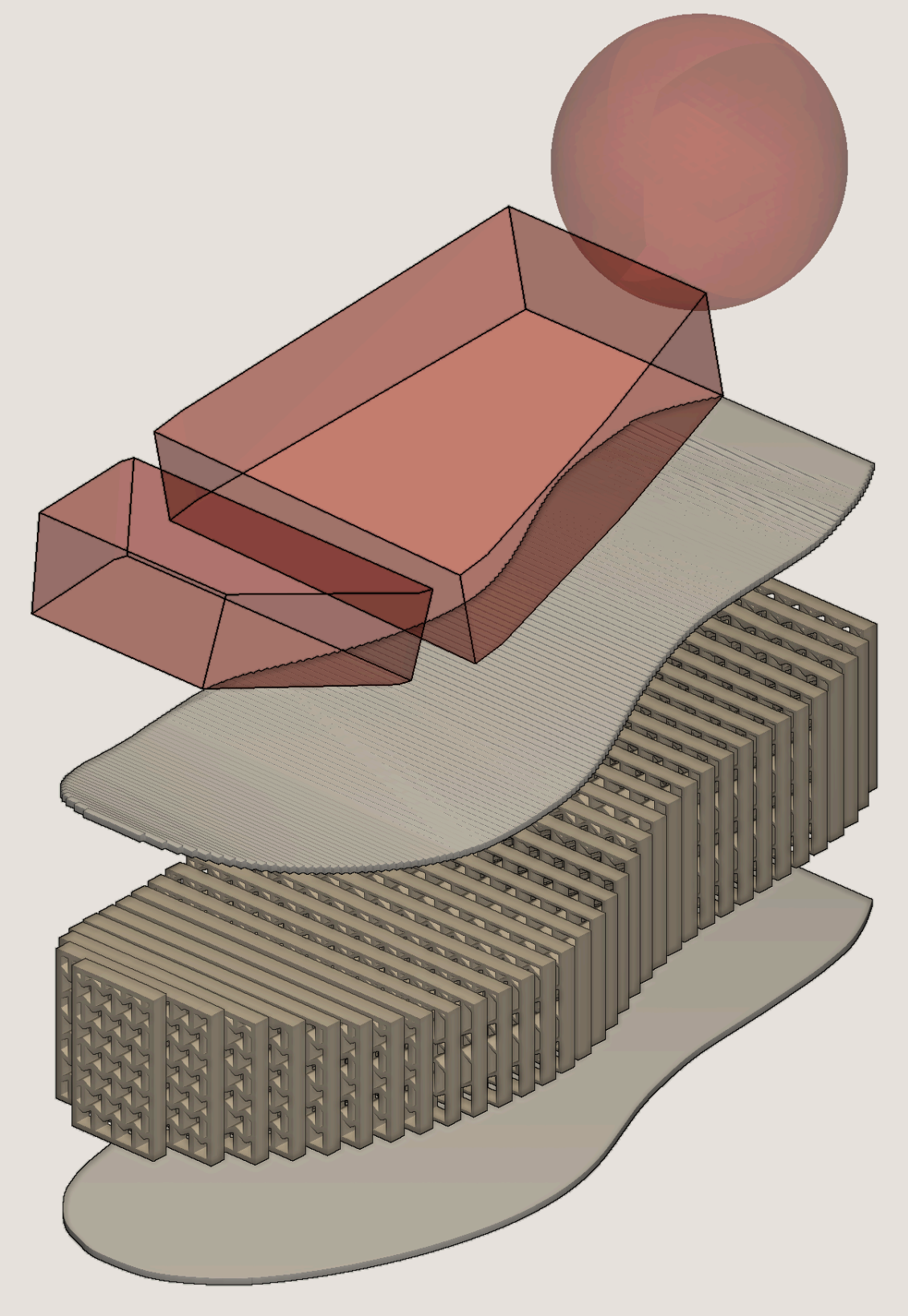


Fig. 24 Exploded assembly: pseudo-foot (red), top plate, 34 zoned slices, bottom plate. One watertight body, 242 cm³.

PLACEHOLDER Slicer preview of the midsole (print orientation, supports)

PLACEHOLDER Photograph of the printed midsole or section

VI

DISCUSSION AND OUTLOOK

6.1 What the models are good for

For printable re-entrant cells the beam models are a starting point, not an answer. Gibson & Ashby and Ohtaki bracket the small-strain stiffness best; Masters & Evans and Malek & Gibson bracket Poisson's ratio best; Ohtaki alone captures the nonlinear rise, but not buckling. Thick walls (t/l = 0.4) push all four beyond their assumptions.

6.2 Limitations

- One runner, a forefoot striker; a heel striker would load the heel very differently.
- The MTP bend comes from the literature, and the heel–midfoot joint is rigid.
- E_s = 19.3 MPa is assumed; the printed TPU will set the true value.
- The 4 × 4 ANSYS block is boundary-affected; periodic conditions would be cleaner.
- Slices interact only through the plates (Winkler, no inter-column shear).
- The zone search used Masters & Evans, which ANSYS shows is too soft at this thickness.

6.3 Next steps

Compression tests of the printed block and a solid TPU coupon will fix E_s and give an experimental curve to set against Figure 11. The loading pipeline is independent of the cell: the chiral, rotating-square and tensegrity strands can be run through the same stance and midsole for a like-for-like comparison. Beyond that, a three-dimensional re-entrant lattice would remove the slice gaps, carry the forefoot flex itself and dome under the foot.

PLACEHOLDER Test set-up and measured force–displacement curve of the printed block

Fig. 27 Experimental validation (to follow).

REFERENCES

[1] Lakes, R. (1987). Foam structures with a negative Poisson's ratio. *Science*, 235, 1038–1040.

[2] Evans, K. E., Nkansah, M. A., Hutchinson, I. J., & Rogers, S. C. (1991). Molecular network design. *Nature*, 353, 124.

[3] Ren, X., Das, R., Tran, P., Ngo, T. D., & Xie, Y. M. (2018). Auxetic metamaterials and structures: a review. *Smart Materials and Structures*, 27, 023001.

[4] Gibson, L. J., & Ashby, M. F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.

[5] Masters, I. G., & Evans, K. E. (1996). Models for the elastic deformation of honeycombs. *Composite Structures*, 35, 403–422.

[6] Malek, S., & Gibson, L. (2015). Effective elastic properties of periodic hexagonal honeycombs. *Mechanics of Materials*, 91, 226–240.

[7] Ohtaki, H., Hu, G., Nagasaka, Y., & Kotosaka, S. (2004). Analysis of negative Poisson's ratio of re-entrant honeycombs. *[verify journal and pages]*

[8] *[Add a footwear-specific auxetic reference — e.g. a study of printed auxetic insoles or midsoles.]*

[9] Stefanyshyn, D. J., & Nigg, B. M. (1997). Mechanical energy contribution of the metatarsophalangeal joint to running and sprinting. *Journal of Biomechanics*, 30, 1081–1085.

[10] Wang et al. (2021). *[MTP joint kinematics in running — PMC8093456; complete the citation]*

[11] Fukuchi, R. K., Fukuchi, C. A., & Duarte, M. (2017). A public dataset of running biomechanics and the effects of running speed on lower extremity kinematics and kinetics. *PeerJ*, 5, e3298.

[12] Wu et al. (2023). *Frontiers in Bioengineering and Biotechnology*, doi:10.3389/fbioe.2023.1251324. *[complete the citation]*

Citations marked in square-bracketed italics need checking against the original sources before submission.